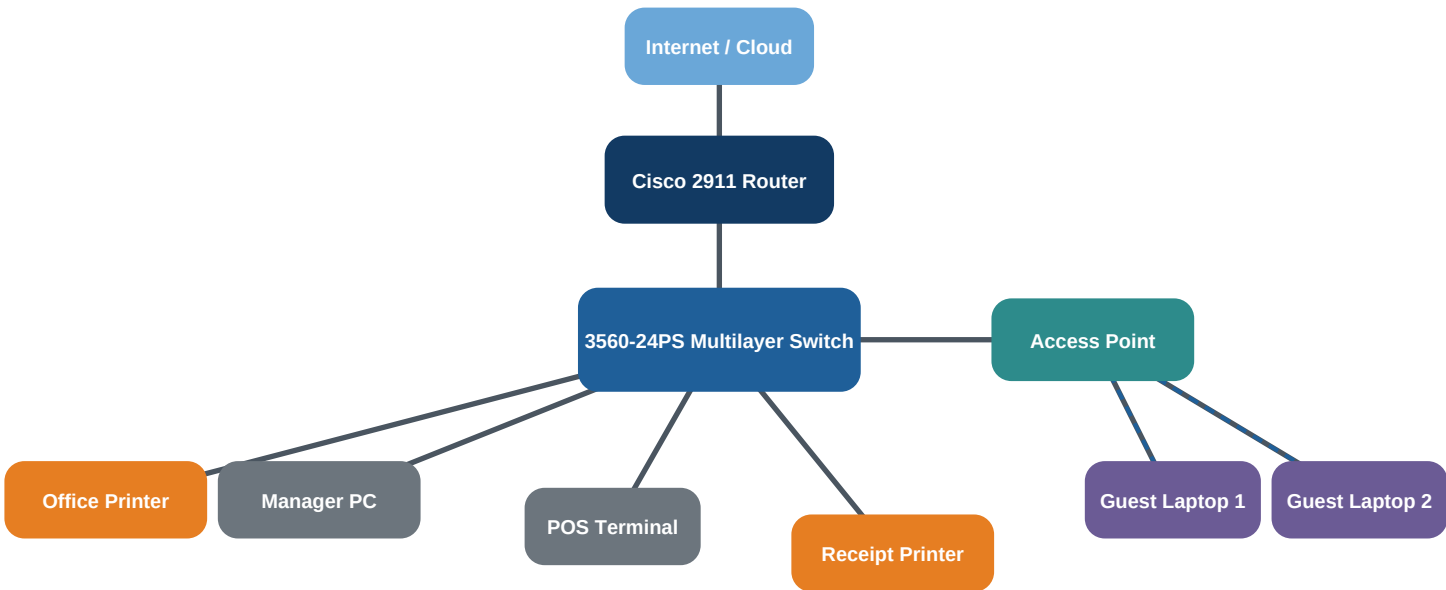


COFFEESHOP NETWORK INFRASTRUCTURE

Cisco Packet Tracer Project Documentation



CoffeeShop segmented network topology

Project objective: Design and validate a structured coffee shop network with departmental VLAN segmentation, Layer 3 inter-VLAN routing, guest wireless access, POS/printer connectivity, and end-to-end connectivity testing.

Project Area	Implementation
Network simulator	Cisco Packet Tracer
Core routing/switching	Cisco 2911 Router + Cisco 3560-24PS Multilayer Switch
Wireless	Access Point + Guest laptops
Segmentation	VLAN 10, 20, 30, 40 and 50
Validation	ICMP ping / 0% packet loss on tested gateways

Prepared as a practical networking project and portfolio documentation.

1. Project Overview

The project models a small coffee shop network in Cisco Packet Tracer. The design separates business functions into logical VLANs so that Management, POS, Staff, Guest Wi-Fi, and Printers can use dedicated IP networks. A Layer 3 multilayer switch provides the VLAN gateway interfaces and inter-VLAN routing, while a Cisco 2911 router provides the upstream/WAN side of the design.

The final topology contains a router, multilayer switch, wireless access point, guest laptops, management PC, POS terminal, and network printers. The project was tested from end devices using the Packet Tracer command prompt and ICMP ping.

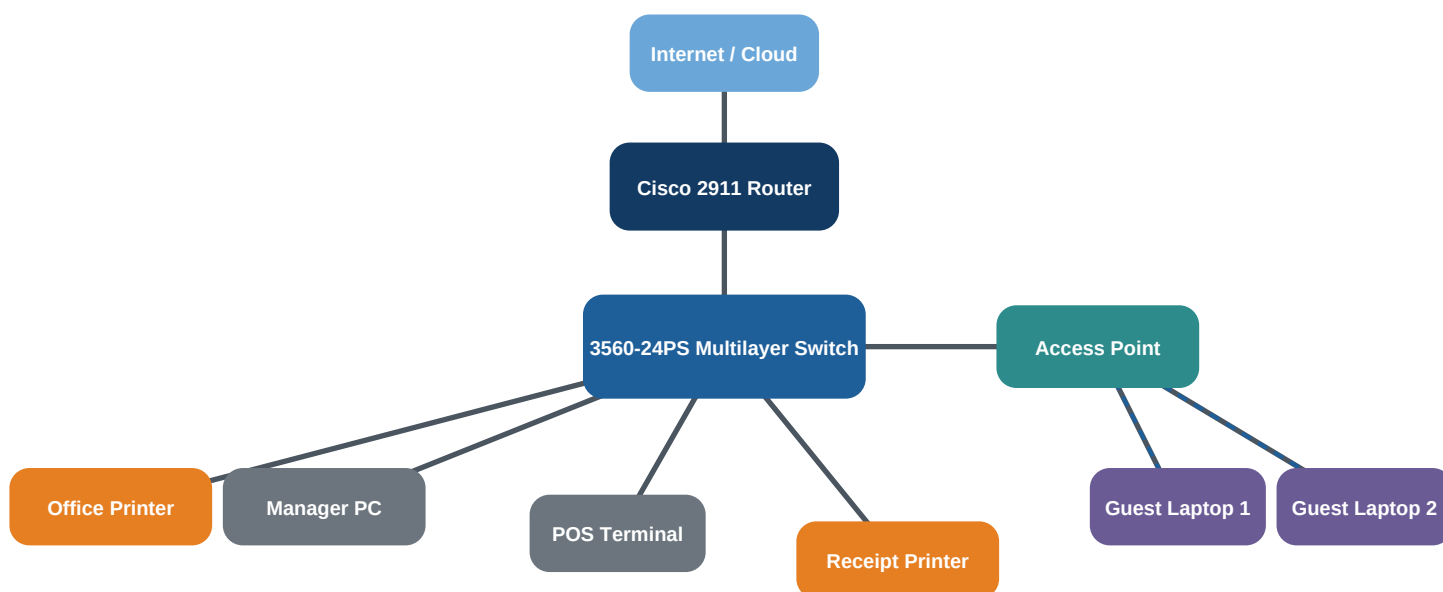
Project goals

- Create a realistic, structured coffee shop network topology.
- Separate departments and services with VLANs.
- Provide Layer 3 gateways and inter-VLAN communication.
- Provide wireless connectivity for guest users.
- Connect POS and printer devices to their intended network segments.
- Verify reachability and troubleshoot connectivity issues.
- Save and document a clean, submission-ready Packet Tracer project.

Final result

The final project was returned to Realtime mode after simulation testing, the temporary PDU/test events were cleared, and the project was saved. The Packet Tracer file was also uploaded to the GitHub repository.

2. Network Topology & Devices



CoffeeShop segmented network topology

Main devices and their roles

Device	Role
Cisco 2911 Router	Upstream routing/WAN connectivity and point-to-point link toward the multilayer switch.
Cisco 3560-24PS Multilayer Switch	Core Layer 3 switch; hosts VLAN interfaces and performs inter-VLAN routing.
Wireless Access Point	Provides Guest Wi-Fi connectivity for wireless laptops.
Manager PC	Management network endpoint (VLAN 10).
POS Terminal	Point-of-sale endpoint (VLAN 20).
Guest Laptop 1 / 2	Guest wireless endpoints (VLAN 40).
Office Printer / Receipt Printer	Printer network endpoints (VLAN 50).
Cloud-PT	Represents the external/internet side of the topology.

3. VLAN & IP Addressing Design

The network uses /24 subnets for each VLAN. The default gateway is the .1 address in each subnet and is configured on the multilayer switch using VLAN interfaces (SVIs).

VLAN	Name	Network	Gateway	Purpose
10	MANAGEMENT	192.168.10.0/24	192.168.10.1	Management devices
20	POS	192.168.20.0/24	192.168.20.1	Point-of-sale system
30	STAFF	192.168.30.0/24	192.168.30.1	Staff operations
40	GUEST_WIFI	192.168.40.0/24	192.168.40.1	Guest wireless access
50	PRINTER	192.168.50.0/24	192.168.50.1	Network printers

Example endpoint addressing observed during testing

Endpoint	IPv4 address	Subnet mask	Gateway
POS Terminal	192.168.20.11	255.255.255.0	192.168.20.1
Guest Laptop 2	192.168.40.11	255.255.255.0	192.168.40.1

The Guest Laptop 2 configuration was verified with **192.168.40.11/24**. Its successful ping to **192.168.40.1** confirmed connectivity to the Guest VLAN gateway.

4. Protocols & OSI Model Layers

Protocol / Technology	Purpose in the project	Primary OSI layer
Ethernet	Wired host-to-switch/router communication.	Layer 1–2
IEEE 802.11	Wireless communication between Guest laptops and Access Point.	Layer 1–2
IEEE 802.1Q / VLAN	Logical segmentation and VLAN tagging on trunk links where applicable	Layer 2
ARP	Resolves IPv4 addresses to MAC addresses on local networks.	Layer 2
IPv4	Logical addressing of VLANs and routed networks.	Layer 3
ICMP (Ping)	Connectivity and troubleshooting tests.	Layer 3
TCP/UDP	Transport protocols used by higher-level network applications.	Layer 4
Routing / Inter-VLAN Routing	Moves packets between different IP networks/VLANs.	Layer 3
STP	Switching-loop prevention technology available at Layer 2.	Layer 2

OSI layer coverage

Layer	Name	How it appears in this project
7	Application	End-user/network applications can operate over the configured IP connectivity.
6	Presentation	Data representation is handled by applications/protocol stacks.
5	Session	Session establishment/maintenance is handled by applications and transport services.
4	Transport	TCP/UDP provide end-to-end transport for applications.
3	Network	IPv4 addressing, gateways and inter-VLAN routing.
2	Data Link	MAC addressing, switching, VLANs and wireless 802.11 framing.
1	Physical	Copper Ethernet links and wireless radio connectivity.

5. Project Implementation Process

The project was built and validated in stages. The following sequence can be used to reproduce the project from an empty Packet Tracer workspace.

Step 1 — Plan the topology

- Identify required departments/services: Management, POS, Staff, Guest Wi-Fi, and Printers.
- Choose the router, Layer 3 multilayer switch, access point, endpoints and printers.
- Define a separate /24 network and gateway for each VLAN.

Step 2 — Place and connect devices

- Place the Cisco 2911 router, Cisco 3560-24PS multilayer switch, Access Point, PCs, laptops and printers.
- Use Copper Straight-Through for appropriate wired Ethernet connections.
- Connect the Access Point to the multilayer switch and associate Guest laptops wirelessly.

Step 3 — Create VLANs

The VLANs were created and assigned names to represent the coffee shop functions.

```
VLAN 10 MANAGEMENT
VLAN 20 POS
VLAN 30 STAFF
VLAN 40 GUEST_WIFI
VLAN 50 PRINTER
```

Step 4 — Assign switch ports

- Assign endpoint-facing switch ports to the correct access VLAN.
- The observed final switch status showed active ports associated with VLANs 10, 20, 30, 40 and 50.
- The Access Point connection was placed in the Guest Wi-Fi segment (VLAN 40).

Step 5 — Configure Layer 3 VLAN gateways

SVIs on the multilayer switch provide the gateway addresses: 192.168.10.1, 192.168.20.1, 192.168.30.1, 192.168.40.1 and 192.168.50.1.

6. Routing & Inter-VLAN Configuration

The multilayer switch is the central Layer 3 device for the VLANs. Its routing table was verified during troubleshooting and showed the VLAN networks as directly connected.

Routing table verification showed these directly connected networks:

```
C 192.168.10.0/24 directly connected, Vlan10  
C 192.168.20.0/24 directly connected, Vlan20  
C 192.168.30.0/24 directly connected, Vlan30  
C 192.168.40.0/24 directly connected, Vlan40  
C 192.168.50.0/24 directly connected, Vlan50
```

The router-to-switch point-to-point network used the 10.0.0.0/30 range. The routing table showed the switch using a default route toward 10.0.0.2, while the router had routes back toward the VLAN networks through 10.0.0.1.

Important troubleshooting lesson

The 10.0.0.1 and 10.0.0.2 addresses belong to the router/switch transit link, not to an end-user VLAN. Therefore, a failed PC ping to those addresses does not automatically mean the VLAN network is broken. The correct validation is to test the local gateway and required destination networks from the relevant endpoint.

7. Guest Wi-Fi Configuration

The Access Point was configured to provide a dedicated guest wireless network.

Setting	Configured value
SSID	CoffeeShop-Guest
Band	2.4 GHz
Channel	6 (as configured during the project)
Security	WPA2-PSK
Encryption	AES
Guest network	VLAN 40 / 192.168.40.0/24

Guest Laptop 2 successfully associated with the Access Point. Its wireless configuration showed IPv4 address **192.168.40.11** with subnet mask **255.255.255.0**.

The guest laptop then successfully pinged **192.168.40.1** with 4 packets sent, 4 received and 0% loss. This confirms the wireless client can reach its VLAN gateway.

8. Testing & Validation

Connectivity was tested from end devices using the Packet Tracer PC command prompt. Successful tests returned replies and 0% packet loss.

Test target	Expected purpose	Result
192.168.20.1	POS VLAN gateway	PASS — 0% packet loss
192.168.30.1	Staff VLAN gateway	PASS — 0% packet loss
192.168.40.1	Guest Wi-Fi gateway	PASS — 0% packet loss
192.168.50.1	Printer VLAN gateway	PASS — 0% packet loss
192.168.10.1	Management VLAN gateway	PASS — 0% packet loss

Example successful test

```
C:\>ping 192.168.40.1

Reply from 192.168.40.1: bytes=32 time=51ms TTL=255
Reply from 192.168.40.1: bytes=32 time=9ms TTL=255
Reply from 192.168.40.1: bytes=32 time=35ms TTL=255
Reply from 192.168.40.1: bytes=32 time=13ms TTL=255

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
```

The project also included troubleshooting of an initial wireless/cabling issue. After reconnecting and correcting the relevant configuration, the guest client associated successfully and the gateway ping passed.

9. Simulation, Troubleshooting & Final Cleanup

Packet Tracer Simulation mode was used to observe packets/events. Red envelope icons in Simulation mode represent PDUs/test packets; they are not themselves an indication that a cable or device is faulty.

Cleanup procedure

- Run the required ping/connectivity tests.
- Use Reset Simulation to clear simulation events when finished testing.
- Remove temporary PDU entries from the PDU list if any remain.
- Return to Realtime mode for the final saved topology.
- Confirm that no temporary red test envelopes remain on the topology.
- Save the final .pkt file.

Final state

The final screenshot showed the topology in Realtime mode with green active links, the Access Point connected to the guest wireless clients, no visible temporary red PDU envelope, and the project saved.

10. GitHub Documentation & Portfolio

The completed Packet Tracer project was uploaded to a public GitHub repository for portfolio use.

Item	Value
GitHub username	SubhanMohammad2244
Repository	coffee-shop-network-cisco-packet-tracer
Project file	Coffeeshop Network project.pkt
Documentation	README.md
Visibility	Public

Suggested LinkedIn summary

Designed and implemented a CoffeeShop Network Infrastructure using Cisco Packet Tracer, featuring VLAN segmentation for Management, POS, Staff, Guest Wi-Fi and Printers, Layer 3 Inter-VLAN routing, wireless connectivity, IPv4 addressing, and ICMP-based validation. Connectivity was verified successfully with 0% packet loss across the tested VLAN gateways.

Suggested LinkedIn hashtags

#Cisco #Networking #CiscoPacketTracer #CCNA #VLAN #InterVLANRouting #NetworkEngineering #Routing #Switching #ITInfrastructure

11. Final Project Summary

This project demonstrates a practical understanding of enterprise-style network segmentation and troubleshooting in a small-business environment. The design separates traffic by function, provides centralized Layer 3 routing, supports wireless guest access, and validates connectivity using standard network troubleshooting methods.

Area	Final status
Topology design	Completed
VLAN segmentation	Completed
Layer 3 SVIs / gateways	Completed
Inter-VLAN routing	Completed
Router transit connectivity	Configured and verified from the network devices
Guest wireless	Connected and gateway tested successfully
POS / printer connectivity	Configured
ICMP validation	Successful; tested gateways showed 0% packet loss
Simulation cleanup	Completed
GitHub upload	Completed
Final Packet Tracer file	Saved

Key skills demonstrated

- Cisco Packet Tracer topology design
- VLAN configuration and segmentation
- Layer 3 switching and inter-VLAN routing
- IPv4 addressing and subnetting
- Wireless network configuration
- ICMP-based connectivity testing
- Network troubleshooting and validation
- Technical documentation and GitHub project publishing

Note: This documentation reflects the configuration and testing steps established during the project work. Some higher-layer protocols listed in the OSI overview describe the networking stack rather than a separately configured application server in Packet Tracer.